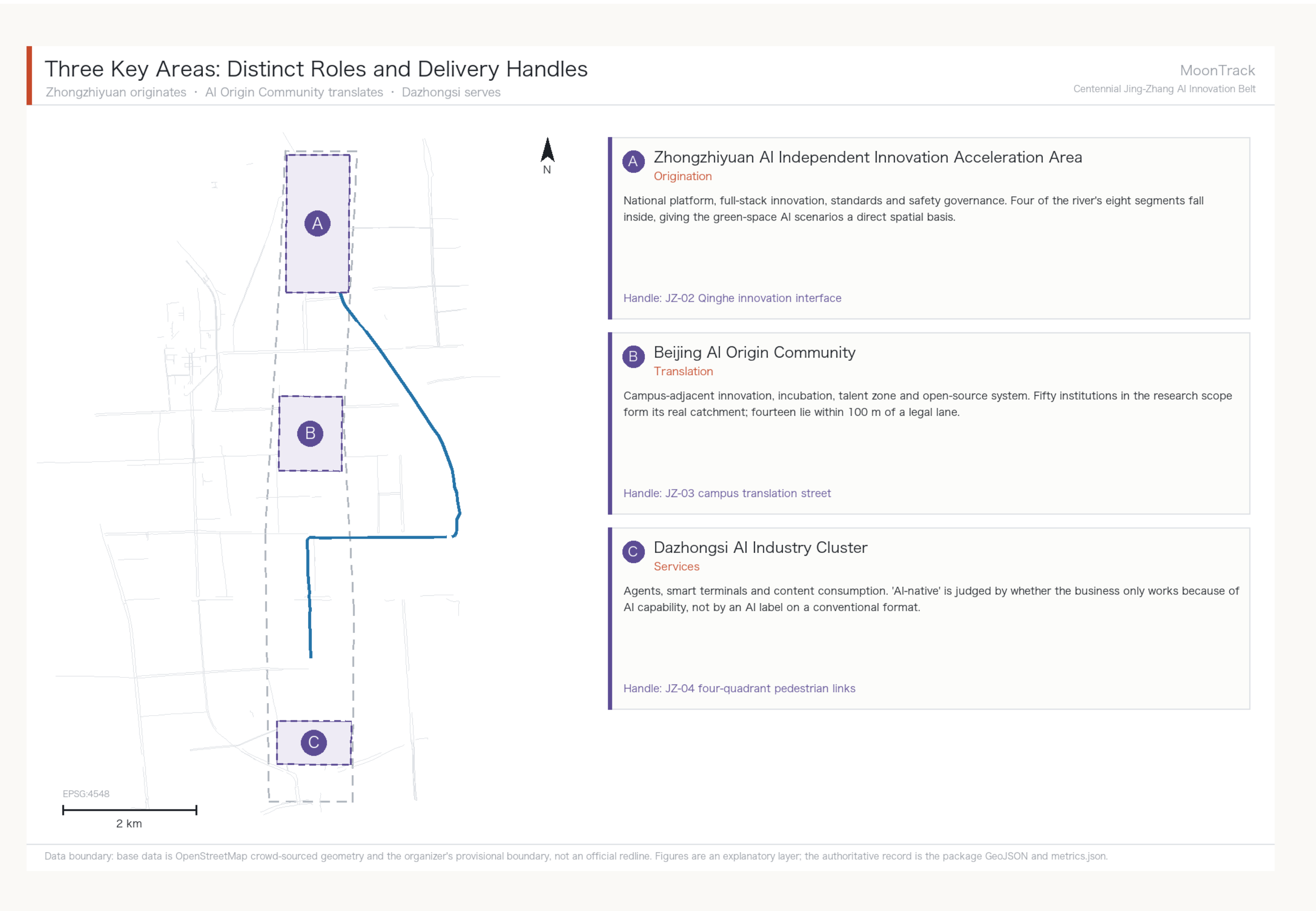
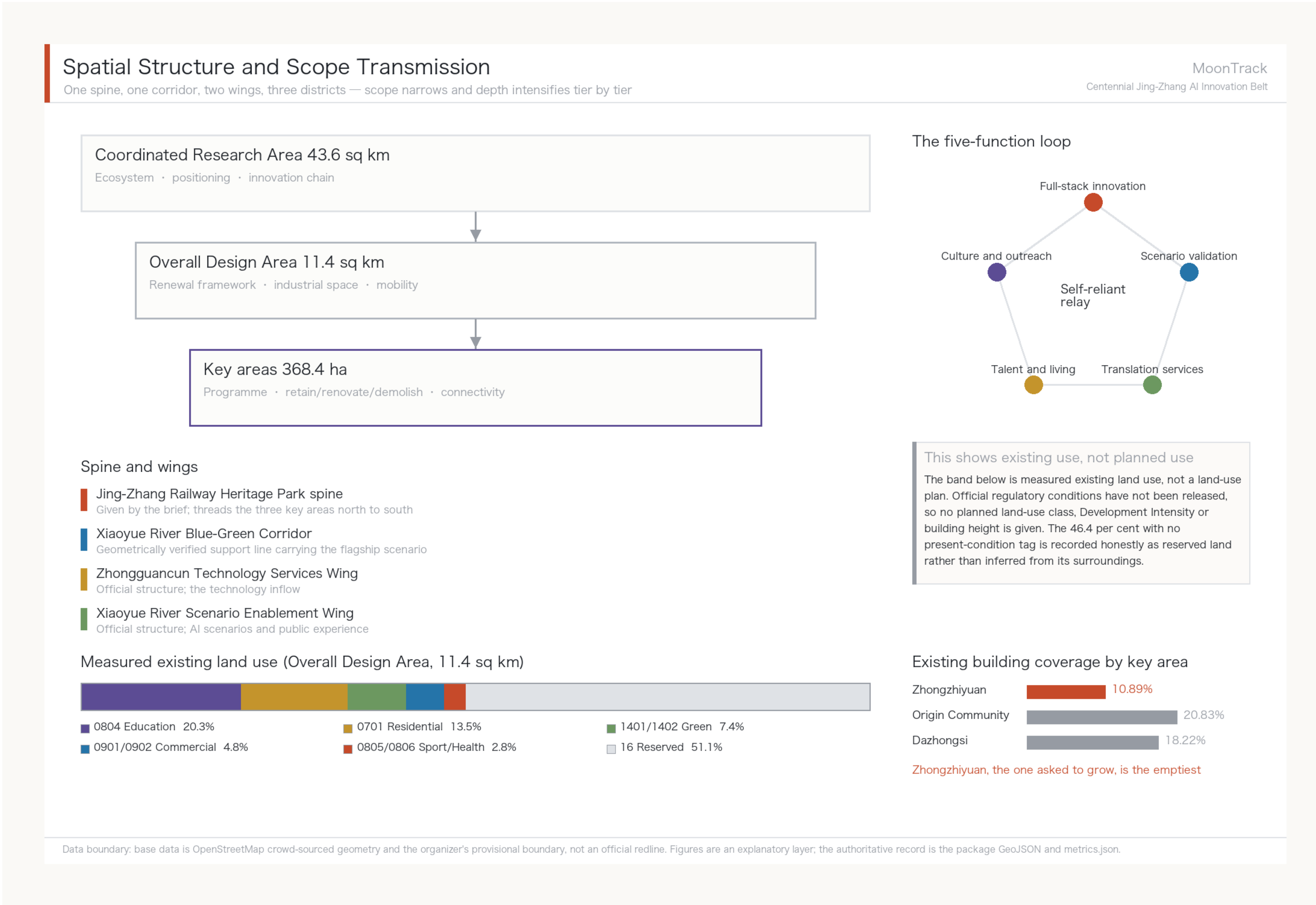
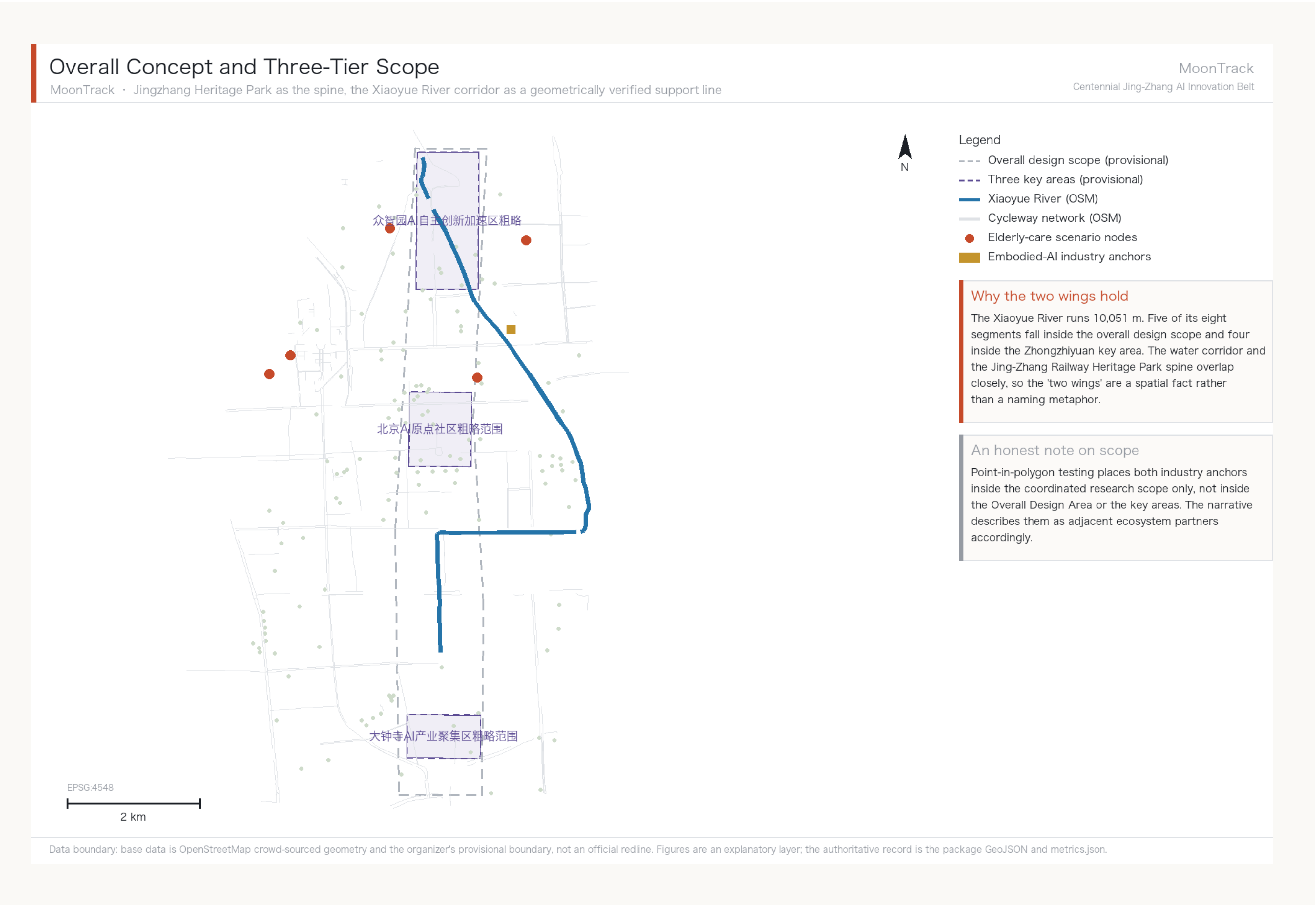


MoonTrack — Concept and Spatial Structure

Centennial Jing-Zhang AI Innovation Belt — Board 1 of 3



Name and concept

'MoonTrack' joins the moon of the Xiaoyue River (literally 'little moon river') to the track of the railway and the platform. It is a concept-level sub-brand and does not replace the official project name. The organising idea is a relay of self-reliant innovation: Zhan Tianyou built the Jingzhang railway without foreign engineers a century ago, which answers the first of the five official functions, a full-stack self-reliant AI system.

How the three tiers transmit

The Coordinated Research Area governs ecosystem and positioning; the Overall Design Area governs the renewal framework, industrial space and mobility; the key areas govern programme, retain-renovate-demolish decisions and public-space connectivity. Scope narrows and depth intensifies tier by tier, and each tier's conclusions hold only within the data available at that tier.

What these sheets do not promise

Official regulatory conditions have not been released, so land-use classes, development intensity, building heights and road redlines cannot properly be drawn and no figures are given for them. The key-area boundaries are the organizer's provisional polygons and cannot support formal scoring or approval.

Cycleway Connectivity: 0 of 5, and 56 Metres

Measured legal right-of-way network for low-speed robots — Board 2 of 3

Cycleway Connectivity and AI Scenario Nodes

Measured legal right-of-way network for low-speed robots: one colour per connected component

MoonTrack

Centennial Jing-Zhang AI Innovation Belt



Legend

- Largest component 27,560 m (14.9%)
- 2nd component 21,580 m
- 3rd component 22,000 m
- 4th component 18,330 m
- Remaining 58 fragments
- Xiaoyue River
- Elderly facility (unreachable)

Core finding: 0 of 5

Network-reachable elderly facilities: zero of five. Four sit in different components from the candidate depot; the fifth is connected but its final 652 m has no legal right of way. Measured as the crow flies all five are within 500 m. The difference is not measurement precision — robots cannot teleport.

56 m unlocks 30 km

The three shortest gaps to neighbouring components measure 9 m, 12 m and 35 m — 56 m in total. If all three are real and passable, the main network grows from 27,560 m to 57,770 m. This sets the phasing: verify the breaks before buying any robots.

What this figure cannot prove

Topological connectivity is not passability. Kerb gradient, paving, clear width and turning radius are invisible here, and gaps of 9 m or 12 m may be OSM mapping artefacts rather than real breaks. JZ-07 is therefore scoped as a field-verification list, not a construction brief.

EPSG:4548

2 km

Data boundary: base data is OpenStreetMap crowd-sourced geometry and the organizer's provisional boundary, not an official redline. Figures are an explanatory layer; the authoritative record is the package GeoJSON and metrics.json.

0 / 5

elderly facilities reachable

62

connected components

14.9%

largest component share

56 m

unlocks 30,210 m

Why this is a design problem, not a technical one

Beijing's interim measures for unmanned delivery vehicles require robots to be managed as non-motorised traffic, to travel within cycleways, and not to exceed 15 km/h. Right of way is fixed by regulation, so 'can it get there' is a computable graph question rather than a metaphor. The answer: under current right of way the flagship scenario cannot run even once.

Direct consequence for phasing

Phase one is not depots and robots; it is verifying and closing three breaks of a few dozen metres. What gets built is a connection in the existing Walking and Cycling Network, not dedicated robot infrastructure — if the robots leave, cyclists and wheelchair users still benefit. This is where the reversibility principle lands.

Evidence boundary

Topological connectivity is not passability; the 8 m snapping tolerance is a method choice; gaps of 9 m or 12 m may be mapping artefacts. These risks are registered as A-NET-001 to A-NET-004.

From Diagnosis to Remedy: Rule, Link, Terminal

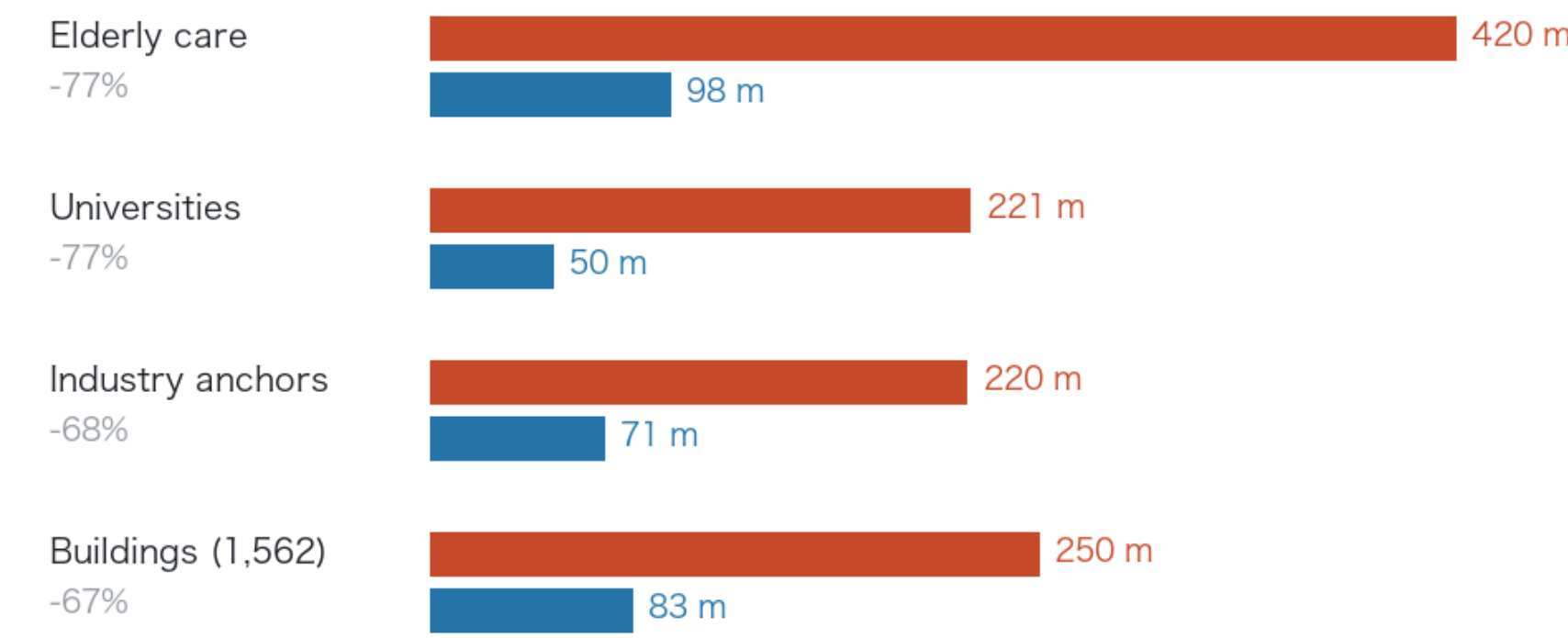
Policy levers and the phase-one deliverable — Board 3 of 3

The Price of the Right-of-Way Rule

The same doors measured against two networks; four interventions, one fixed depot

Median terminal distance, same doors

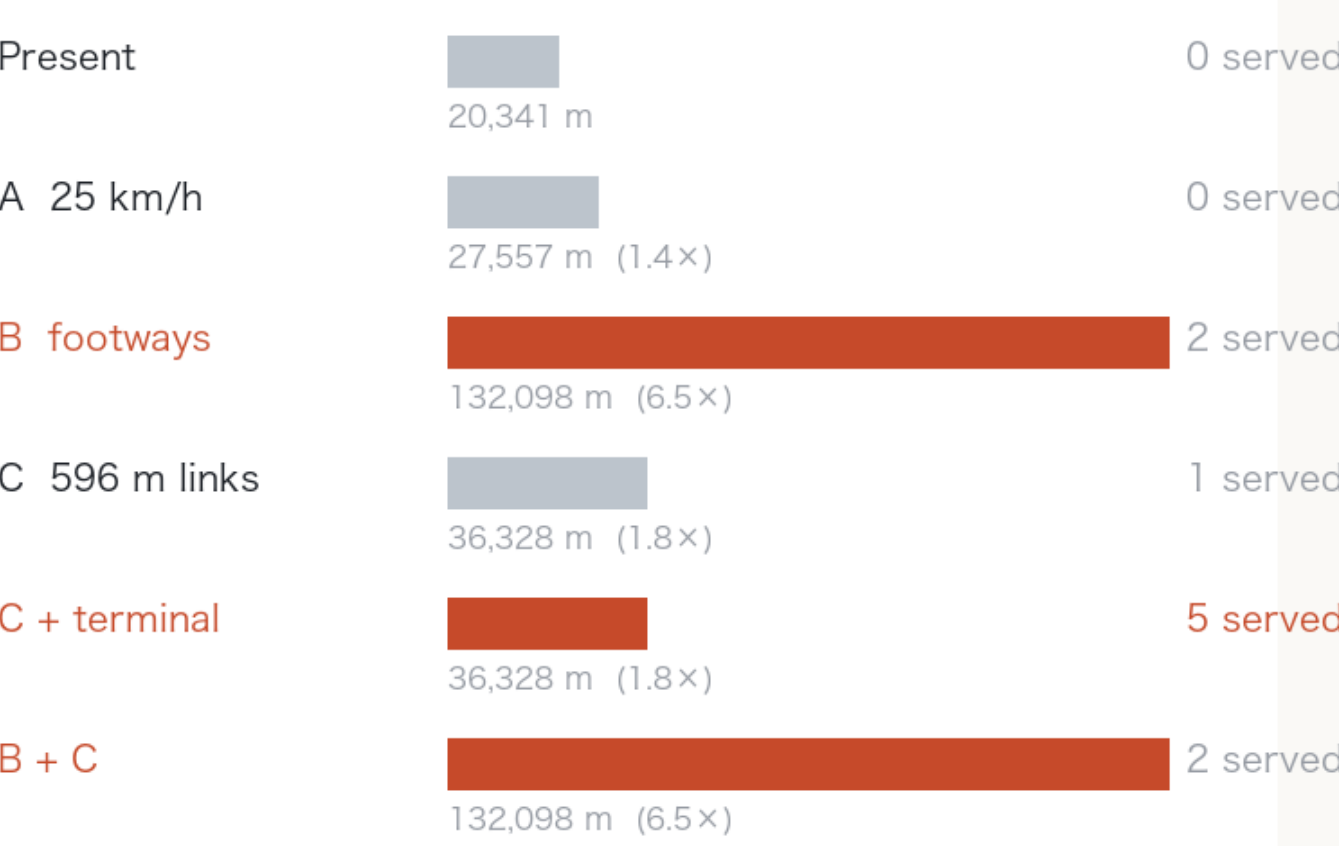
Upper bar: cycle lanes only (present rule). Lower bar: footways permitted



This column is the service cost of that rule
The present measures require delivery robots to stay in cycle lanes. For the same doors the median terminal distance is tens of metres against the walking network and two to three hundred against legal right of way; the rule multiplies it three- to fourfold. Steps are excluded, since neither a robot nor a wheelchair can climb them.

Four interventions, which one works

Test: terminal ≤150 m and within 15 min of legal travel; depot held fixed



The order is computed, not asserted
Raising the speed limit serves no additional facility. Changing the right-of-way rule costs nothing to build and multiplies the reachable network 6.5 times. The 596 m of links give 1.79. Only links plus terminal works serve all five. Rule first, then links, then terminal works. Footway mapping is patchy, so 6.5x is a lower bound.

Data boundary: base data is OpenStreetMap crowd-sourced geometry and the organizer's provisional boundary, not an official redline. Figures are an explanatory layer; the authoritative record is the package GeoJSON and metrics.json.

6.5x

network gained by permitting footways

0/10

deliveries under today's rules

596 m

of link to join all five

30%

ledger audit completeness

The sequence is computed, not asserted

Change the rule first: permit low-speed devices on footways (steps excluded). No construction, and the 15-minute network goes from 20,341 m to 132,098 m. Then close the links: 596 m joins all five facilities into one network. Then do the terminals: 1,841 m to reach the last four doors. Raising the speed cap is void - from 15 to 25 km/h serves not one additional facility.

What the 30-task ledger says

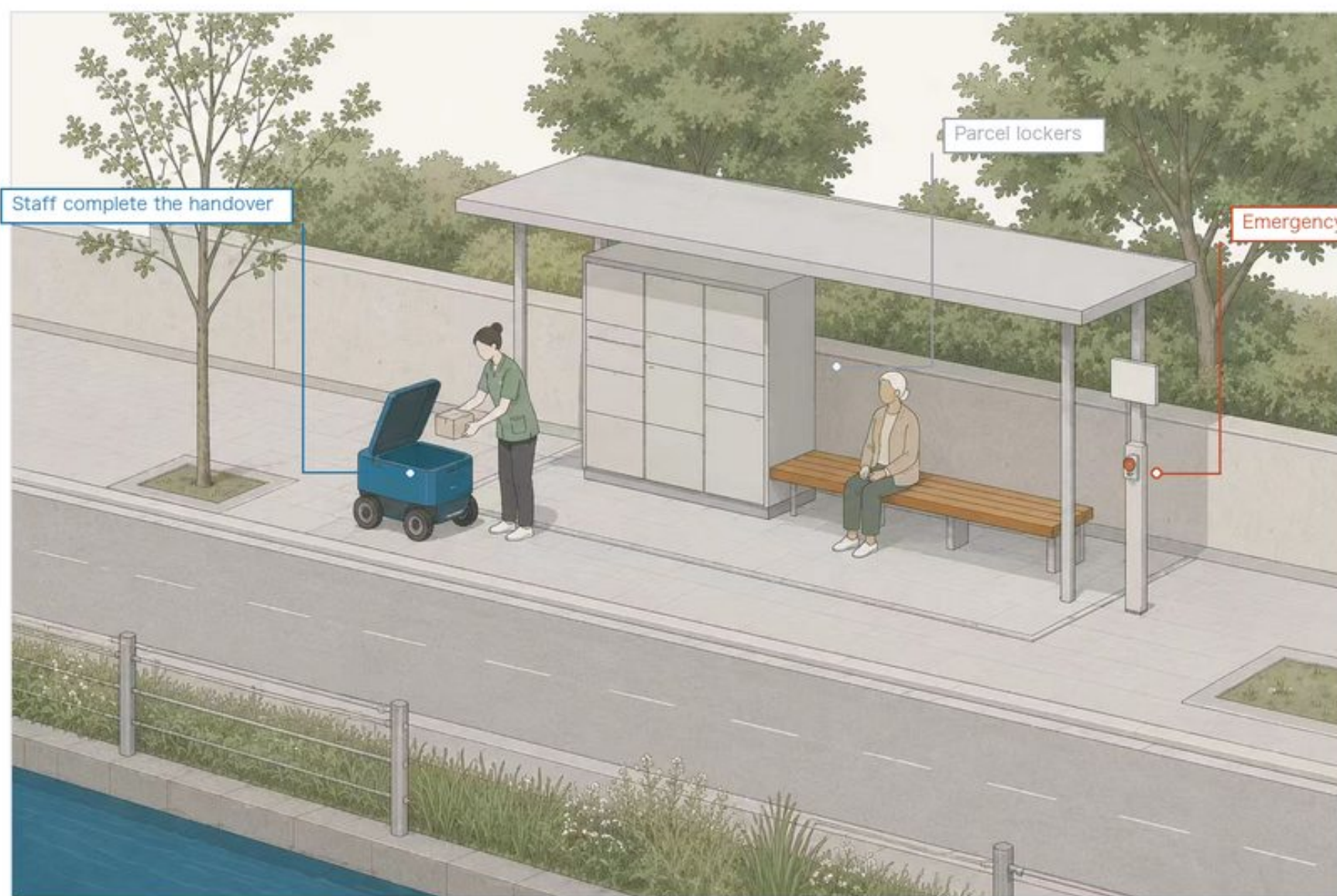
Ten real destinations across three right-of-way states, every outcome decided by Dijkstra on the measured graph rather than by synthetic trajectories. Today: 0 of 10. After the 596 m, network reachability rises from 2 to 6 but deliveries only from 0 to 1 - the construction lands and tasks are still blocked at the door. For the same doors, a person on foot faces 82.8 m against the robot's 249.6 m.

The handover boundary drives the component

No entry indoors, no contact with the resident, no automated decisions. The cabinet therefore opens only to the street and has no indoor-side aperture - the boundary is not a sentence in the notes, it is why the object has this shape. Phase one: 82 m of link, one facility of five, 14.0 minutes to the furthest once joined. The illustration is an AI-generated concept image, not a photograph.

The Platform Unit — Phase One Plate

Phase one is not all five connected: it is 82 m, one stop, one facility



The illustration is an AI-generated conceptual expression, not a site photograph, and depicts no existing place.

This stop is a computed result

Phase-one link	82 m
Facilities served	1 / 5
Worst run once merged	14.0 min
Terminal access, five sites	1,950 m

The handover boundary is built into the component

The robot stops at the door and staff take delivery: it does not enter the building, does not interact with the resident directly, and makes no automated decision. Those three rules are not a line in an operations manual — they are why the unit is shaped this way, opening only to the street side with no opening on the building side.

Why phase one serves only one

Terminal distances for the five facilities are 109, 218, 420, 551 and 652 m. Drawing the line at 150 m, the distance at which a round trip to collect a delivery does not become a burden, only one qualifies. The remaining 1,841 m across four sites is walking and accessibility work, separately scoped and scheduled. Writing phase one as "all five connected" would look better and could not be built.

Data boundary: base data is OpenStreetMap crowd-sourced geometry and the organizer's provisional boundary, not an official redline. Figures are an explanatory layer; the authoritative record is the package GeoJSON and metrics.json.